Assessing Digital Data Storage in Microbial Genomes Through Statistical Simulation

***Abstract*—DNA can carry digital information in a compact form. When that information is placed inside a living microbe, it can also be copied as the organism reproduces. The difficulty is keeping the stored sequence unchanged. Mutations, random changes in a population, and differences in growth can gradually make the data unreadable. This paper describes a simulation that examines these effects together. A 4 kB file is divided into indexed DNA fragments with error-detection fields, stored with different levels of redundancy, and followed through a model of microbial evolution. Recovery is estimated by sampling reads and applying a simplified consensus decoder. The reported experiments use 200 repetitions and mutation rates between 10^{-7} and 10^{-5} per base per generation. Extra copies improve recovery when the payload is neutral, but their benefit is much smaller when damaged variants have a growth advantage. Increasing sequencing depth helps at low coverage, although it cannot restore a sequence that has already disappeared. The results describe the behaviour of the model; they do not establish storage lifetimes for an engineered organism.**

***Index Terms*— DNA storage; microbial genomes; simulation; mutation; genetic drift; redundancy; data recovery.**

I. INTRODUCTION

An archive is useful only if its contents remain readable. Conventional storage systems need maintenance, replacement hardware, and occasional transfers to newer formats. DNA offers another way to represent information: digital symbols can be mapped to nucleotide sequences and later reconstructed from sequencing reads. Its potential density makes it attractive for data that are kept for long periods and accessed infrequently.

Most DNA storage approaches treat the molecules as material kept outside living cells. Genomic storage adds a different possibility. Once an encoded sequence is introduced into a microbial genome, cell division can produce further copies. This makes replication part of the storage process, but it also brings biological change into the system. A descendant may carry a damaged version of the original message, and a variant that grows faster may become common even when it no longer preserves the data.

The practical question is therefore how much of a file can be recovered after a given number of generations. This study examines the roles of mutation, redundancy, selection, and sequencing coverage. It uses a population model to explore their combined effects before attempting biological validation.

II. BACKGROUND AND MOTIVATION

Cell-free storage methods use techniques such as overlapping fragments, error-correcting codes, constrained sequence design, and protective packaging. Storage in living organisms must address another problem: the carrier changes while it makes copies. Earlier demonstrations show that structured information can be placed in cellular DNA, but demonstrating storage is different from establishing reliable recovery over many generations.

A simulation makes it possible to vary individual assumptions and compare outcomes under controlled conditions. Here, a neutral population model provides a useful baseline, while selection tests how strongly a small growth advantage for damaged variants can change the result.

III. STORAGE MODEL AND ENCODING

The workflow has four stages: convert the file into DNA fragments, assign copies to storage locations, propagate the population, and attempt recovery at fixed checkpoints. It separates inherited damage from errors introduced during sequencing. This distinction matters because deeper sequencing may reduce sampling uncertainty, whereas it cannot recreate an original sequence that has been lost from the population.

A. Fragment structure

The file is divided into blocks of 160 bits. Each block occupies an 80-nucleotide payload field under a two-bits-per-base mapping. A 12-nucleotide index records the block position, and an 8-nucleotide cyclic redundancy check (CRC) provides error detection. Each complete fragment therefore contains 100 nucleotides. With 200 fragments, the payload is 32,000 bits, or 4,000 bytes.

TABLE I
FRAGMENT STRUCTURE

Field	Length
Index	12 nucleotides
Payload	80 nucleotides (160 bits)
CRC	8 nucleotides
Total	100 nucleotides

This mapping is used for density calculations. It does not enforce limits on repeated bases or other sequence constraints that may be needed in practice. The resulting density should therefore be read as an idealised value. The CRC detects errors in the model; it is not a mechanism for

correcting a damaged sequence.

B. Multiple copies

Each fragment is stored at R locations, with R set to 1, 2, 3, or 4. Recovery succeeds if at least one copy produces an accepted, correct fragment. Repetition reduces the chance of losing a block, but it also increases the number of inserted bases. The simulation treats the mutation processes of different copies as independent. Real genomic copies can have linked histories and can be affected together by large deletions or recombination, so independence is an approximation.

C. Mutation probability

Let u denote the per-base probability of a damaging change in one generation, and let L be the fragment length. The chance that at least one base in a fragment changes is:

$$q = 1 - (1 - u)^L \approx Lu \quad (1)$$

The approximation is useful when Lu is small. In this model, any damaging change moves a fragment from an intact state into a mutated state. Reverse changes that restore the exact original fragment are omitted. Insertions and deletions are not represented explicitly; their possible effects are absorbed into the overall damage probability. Consequently, the model tracks whether a fragment is intact rather than the detailed sequence of every mutant.

D. Selection and genetic drift

For each fragment copy, x of the P cells carry a mutated variant, giving a mutant frequency $p = x/P$. Intact variants have relative fitness 1 and mutated variants have fitness $1 + s$. A positive s therefore represents an advantage assigned to the mutated state. The selection step is followed by mutation:

$$p_{sel} = p(1 + s) / [p(1 + s) + 1 - p] \quad (2)$$

$$p_{next} = p_{sel} + (1 - p_{sel})q \quad (3)$$

The next generation is drawn as $x_{next} \sim \text{Binomial}(P, p_{next})$. This sampling step represents genetic drift: even with the same expected frequency, chance produces different outcomes across repetitions. The setting $s = 0.01$ gives every damaged state a 1% fitness advantage in the model. It does not mean that every real mutation would remove an equivalent biological burden.

IV. READING THE DATA AND MEASURING RECOVERY

A. Sampling and consensus

At each checkpoint, the number of reads for a fragment copy is sampled from a Poisson distribution with mean C. A read is drawn from the mutated part of the population with probability p. With a per-base sequencing error probability e, the chance of reading a complete L-base fragment without a sequencing error is $(1 - e)^L$.

The simplified decoder compares correctly read intact fragments with correctly read mutant fragments. It accepts the intact sequence only when the intact count is strictly larger. A tie or failure is treated as an erasure. Random sequencing errors are assumed not to form a competing consensus, while reads assigned to a mutant lineage are assumed to agree. This is a coarse model of decoding, rather than a full base-by-base reconstruction algorithm. A real CRC also has a finite probability of accepting an erroneous sequence, which is not included here.

B. Monte Carlo evaluation

Each parameter configuration is repeated 200 times using separate random streams. At every checkpoint, the simulation records both the proportion of recovered fragments and whether all 200 fragments were recovered. Repeating the experiment captures variation caused by mutation, drift, and read sampling. The source study describes a seeded NumPy implementation that processes repetitions in vectorised arrays; the code, seed values, and raw outputs were not supplied with the paper, so its numerical results are retained here as reported.

C. Experimental settings

TABLE II
SIMULATION PARAMETERS

Parameter	Values
Fragment length / fragments per file	100 nucleotides / 200
Redundancy R	1, 2, 3, 4
Population size P	1,000 cells
Mutation probability u	10^{-7}, 10^{-6}, 10^{-5} per base per generation
Selection coefficient s	0 or 0.01
Sequencing error probability e	0.005 per base
Mean coverage C	5, 10, 30, 100 reads per fragment copy
Simulation horizon / checkpoints	3,000 generations / every 150 generations
Repetitions	200 per configuration

The mutation rates were chosen to make deterioration visible within the simulation horizon. They represent an elevated-rate regime rather than a general estimate for ordinary microbial growth. The coverage comparison uses $u = 10^{-6}$ and $s = 0.01$. Conclusions about much lower mutation rates require extrapolation and depend on whether the model remains appropriate.

D. Evaluation measures

Recovery accuracy is the fraction of fragments reconstructed correctly, averaged over repetitions. Because each fragment carries the same amount of data, this also represents the recovered payload fraction. **Full-file recovery probability** is the fraction of repetitions in which every fragment is recovered; it is a more demanding measure.

Nominal information density is $D = 160/(100R) = 1.6/R$ bits per inserted base. If A is the recovered fraction, recoverable density at that checkpoint is $A \times D$. **Storage threshold time** is the first checkpoint at which mean

accuracy falls below 0.99. Since measurements are 150 generations apart, it identifies a failure checkpoint rather than an exact crossing time.

V. RESULTS: NEUTRAL STORAGE AND REDUNDANCY

A. Comparison with a neutral benchmark

In the neutral model, a new mutant has fixation probability 1/P. Combining this with the expected introduction of Pq new mutants gives q as a long-run substitution-rate benchmark. The study compares simulated recovery with exp(-qg), where g is the generation count. This is a useful reference curve, but finite-time consensus recovery also depends on mixed populations and read sampling, so it need not equal a fixation-based prediction.

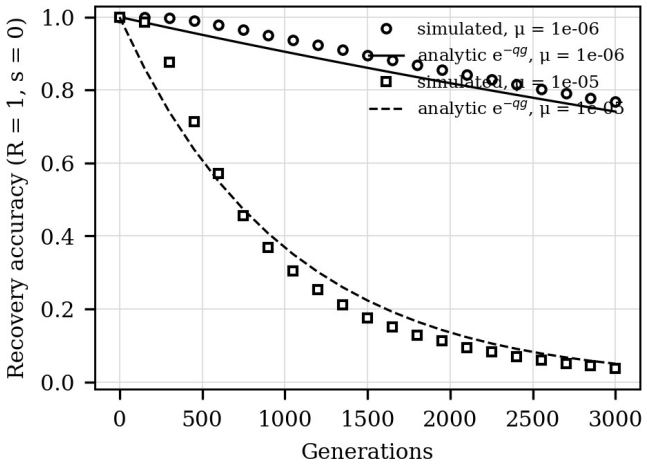


Fig. 1. Neutral-model comparison reproduced from the supplied simulation paper.

TABLE III
NEUTRAL BENCHMARK COMPARISON

Mutation rate	Generation	Reported accuracy	exp(-qg)
10^{-6}	1,500	0.895	0.861
10^{-6}	3,000	0.767	0.741
10^{-5}	1,500	0.176	0.223
10^{-5}	3,000	0.038	0.050

The listed absolute differences are below 0.05. The comparison is consistent with the expected downward trend, but it does not by itself establish complete validation of the simulator. Transient population behaviour and Monte Carlo variation may contribute to the differences; their separate effects cannot be determined from these summary values.

B. Benefits of storing extra copies

At u = 10^{-6} without selection, a single copy gives reported accuracy of 0.950 after 900 generations and 0.767 after 3,000. At 3,000 generations, three copies raise accuracy to 0.988 and four copies to 0.997. Repetition is helpful because losing one copy does not necessarily remove the last intact version of a fragment.

Full-file recovery reveals a stricter requirement. At u = 10^{-7} and generation 900, one copy recovers the complete file in 46% of repetitions, while two copies do so in every reported repetition. At generation 3,000, the reported full-file recovery rates for one through four copies are 1%, 90%, 99%, and 100%. A value of 100% means no failures were observed among the 200 repetitions; it is not a guarantee for future trials.

VI. RESULTS: SELECTION AND SEQUENCING DEPTH

A. What changes when damaged variants grow faster?

Selection produces a much steeper decline. With u = 10^{-6} and s = 0.01, the reported single-copy accuracy is 0.926 after 300 generations, 0.328 after 900, and 0.006 after 3,000. Four copies reach only 0.022 at generation 3,000. Additional copies delay loss, but they do not remove the advantage given to mutated variants.

The study uses the approximation 2s for the fixation probability of a favourable new mutant. Relative to the neutral value 1/P, this suggests an increase of about 2sP, or 20 for the chosen parameters. This is a heuristic explanation of the trend, not an exact prediction for every simulated trajectory. Its relevance to an actual strain would depend on how mutations affect growth.

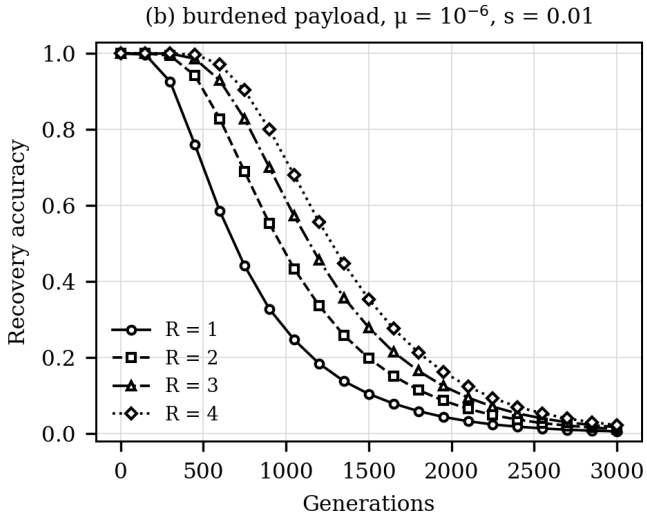


Fig. 2. Reported recovery under selection favouring mutated variants.

B. How much sequencing is useful?

The coverage experiment measures accuracy at generation 300, with u = 10^{-6} and s = 0.01. Increasing coverage reduces uncertainty when few reads are available. The benefit becomes smaller once the sampled population itself contains substantial inherited damage.

TABLE IV
RECOVERY AT DIFFERENT SEQUENCING DEPTHS

Mean reads per copy	R = 1 accuracy	R = 2 accuracy
5	0.833	0.973
10	0.899	0.990
30	0.926	0.994
100	0.933	0.996

For two copies, ten reads per copy reach mean accuracy of 0.990 in this setting. A single copy reaches only 0.899 at the same depth, so ten reads should not be treated as a universal requirement for reliable storage. Moving from 30 to 100 reads produces relatively little improvement in either case. More reads help the decoder observe what remains; they cannot recover an original sequence that is absent.

VII. STORAGE DURATION AND DENSITY

A. First checkpoint below 99% mean recovery

The following values use coverage C = 30. Each number marks the first measured checkpoint at which mean accuracy drops below 0.99. A value greater than 3,000 means the threshold was not crossed within the simulation, rather than that an exact lifetime beyond 3,000 generations was established.

TABLE V
FIRST CHECKPOINT BELOW 99% MEAN RECOVERY

Mutation rate	s	R=1	R=2	R=3	R=4
10^{-7}	0	1,650	>3,000	>3,000	>3,000
10^{-7}	0.01	450	900	1,650	2,250
10^{-6}	0	450	1,500	2,850	>3,000
10^{-6}	0.01	300	450	450	600
10^{-5}	0	150	300	450	600
10^{-5}	0.01	150	150	300	300

Both mutation and selection change the time scale substantially. At the lowest tested rate, a neutral single copy first falls below the threshold at generation 1,650, while two or more copies remain above it throughout the run. Giving mutated variants a 1% advantage moves the single-copy failure checkpoint to generation 450. Four copies extend that checkpoint to generation 2,250, but still do not preserve the same behaviour as a neutral payload.

At $u = 10^{-5}$, every tested configuration crosses the threshold by generation 600. This illustrates the limits of repetition: redundancy can delay deterioration, but it cannot compensate indefinitely for a high rate of inherited damage.

B. Extrapolating to lower mutation rates

The source paper also considers $u \approx 2 \times 10^{-10}$. For a 100-base fragment, q is then approximately 2×10^{-8}. Under the exponential reference model, the characteristic time 1/q is approximately 5×10^{7} generations. This is an analytical extrapolation, not a simulated observation or a guarantee of complete-file recovery.

The characteristic time must also be distinguished from the 99% threshold. At g = 1/q, the reference curve has fallen to exp(-1), about 0.368. Thus, a large characteristic time does not mean that the complete file remains almost perfectly recoverable for that duration. Translating generations into elapsed time would additionally require a specified growth and storage regime.

C. The cost of redundancy

TABLE VI
DENSITY WITH MULTIPLE COPIES

Number of copies R	Nominal payload density
1	1.60 bits per inserted base
2	0.80 bits per inserted base
3	0.53 bits per inserted base
4	0.40 bits per inserted base

More copies exchange density for reliability. The useful choice depends on the required recovery level and storage duration. In the reported neutral cases, moving from one copy to two provides a substantial improvement. Under selection, even four copies may be inadequate. A design should therefore consider the biological effect of the inserted sequence alongside its coding scheme. Maximising the number of copies alone is not enough.

VIII. LIMITATIONS AND INTERPRETATION

The model deliberately reduces a complex biological system to a small set of states. This makes the experiment manageable, but it also limits the conclusions. The results are most useful for comparing assumptions within the model and identifying questions for later experiments.

A. Damage and decoding are simplified

All damaging changes are grouped into one mutated state. Actual substitutions, insertions, deletions, and rearrangements can affect a payload differently. Some may be corrected by a suitable code, while others can destroy the index or several adjacent fragments. Treating every changed fragment as unusable can be pessimistic in some situations. On the other hand, assuming perfect error detection and uncomplicated consensus can be optimistic. The overall bias cannot be described by a single direction.

B. Genomic copies may fail together

Separate copies are treated as independent mutation processes, although copies inside the same genome share inheritance and can be affected by linked events. Recombination between repeated sequences, large deletions, or a population bottleneck could remove several copies together. Such events would reduce the protection expected from independent repetition.

C. Population size and fitness need further study

The population contains 1,000 cells throughout each run. Larger populations can have different transient behaviour even when a neutral long-run substitution-rate benchmark is unchanged. The model also gives the same fitness advantage to all mutated states. In a real organism, many mutations would not relieve the cost of carrying a payload, and some could make growth worse. A more detailed fitness model would be needed before interpreting the selection setting as a measured biological burden.

D. Density and lifetime estimates are conditional

The two-bit mapping leaves out sequence constraints and any additional coding overhead required for synthesis and recovery. The reported density values therefore do not describe a complete physical storage system. Likewise, the elevated mutation-rate experiments do not directly establish long-term behaviour at ordinary rates. Horizontal gene transfer, handling losses, fluctuating growth conditions, and laboratory bottlenecks are outside the model.

E. Reproducibility and biological validation

Independent reproduction would require the simulator, exact random seeds, decoder implementation, and configuration files. Raw outputs would also allow confidence intervals and variability to be assessed. The present results have not been validated against sequencing measurements from an engineered strain. Such validation is necessary before making practical reliability claims.

IX. CONCLUSION AND FUTURE WORK

The simulation shows how microbial data storage can be studied as a recovery problem shaped by mutation, drift, selection, and sampling. Within the tested settings, repeated copies improve the survival of neutral payloads. A growth advantage for damaged variants sharply reduces that benefit. Sequencing depth helps when reads are scarce, but its value becomes limited once inherited damage is widespread.

A useful next version would represent explicit sequence changes, linked copies, realistic fitness effects, and population bottlenecks. It could also compare repetition with an outer error-correcting code and incorporate sequence-design constraints. Most importantly, its predictions should be compared with observed recovery from biological samples. Until then, the framework provides a way to examine design trade-offs rather than a demonstrated guarantee of durable genomic storage.